

Lee Yao

DATA ENGINEER · ANALYTICS ENGINEERING · AI SOLUTIONS

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SUMMARY

Data Engineer with experience building cloud-native data platforms, analytics pipelines, and AI-enabled data solutions. Skilled in designing end-to-end ELT workflows using **Python, Airflow, dbt, Snowflake, AWS, Spark, and Kafka**. Experienced in dimensional modeling, data quality frameworks, and integrating LLM-powered analytics into modern data platforms. Designed and implemented data platforms for healthcare, financial, e-commerce, and transportation domains using real-world datasets and production-inspired architectures.

SKILLS

DATA ENGINEERING	ELT / ETL Pipelines(Python, SQL) · Orchestration (Airflow, Prefect) · Transformation (dbt) · Spark · Kafka · Snowflake · Cloud Ingestion (AWS S3) · Docker
DATA MODELING	Kimball Methodology · Star Schema · Fact & Dimension Modeling · Surrogate Keys· Data Marts
DATABASES	PostgreSQL · Snowflake · Neo4j (Graph) · DuckDB
PROGRAMMING	SQL · Python (Pandas, Matplotlib)
FULL-STACK	End-to-end web application development — frontend, backend, APIs, and deployment
BI & ANALYTICS	Power BI · Qlik Sense
TOOLING	Docker · Git / GitHub · Power Automate · Power Apps · Excel
AI & ANALYTICS	Snowflake Cortex · OpenAI API · LLM-powered Analytics · Prompt Engineering · Power BI · Qlik Sense

EXPERIENCE

Solvía One LLC — Co-Founder & Data Platform Engineer

Full-time

May 2025 – Present

- Designed and deployed the company's website, data infrastructure, and operational automation systems supporting a growing e-commerce business.
- Built cloud-based data workflows and reporting processes supporting product management, customer analytics, and business operations.
- Integrated LLM-powered content generation and workflow automation using OpenAI APIs and modern automation tools to improve operational efficiency.
- Owned technical architecture, implementation, deployment, and platform maintenance as the primary technical contributor.

Freelance / Independent — Data Engineering & Analytics

Queens, NY

Jun 2024 – Present

- Designed and delivered **4 end-to-end data engineering pipelines** on real-world datasets (NYC Taxi, Olist e-commerce, US DOT flights & CMS Medicare), spanning cloud ingestion (AWS S3), warehouse modeling (Snowflake, PostgreSQL), transformation (dbt), orchestration (Airflow, Prefect), and BI reporting (Power BI).
- Built Kimball dimensional models and automated dbt data-quality tests across staging / core / mart layers; orchestrated parameterized, idempotent DAGs supporting monthly backfills with no manual intervention.
- Applied **Snowflake Cortex LLM** for multilingual sentiment analysis on 100K+ records and **Neo4j** graph modeling for fraud-cluster detection.

SDG Group USA — Data Analyst

Bedminster, NJ

Dec 2022 – May 2024

- Developed and maintained **Qlik Sense and Power BI** dashboards for client reporting, operational monitoring, and decision-making.
- Performed data cleaning, validation, and reconciliation to improve accuracy and consistency across reporting views.
- Supported a BI improvement initiative for an airport-retail client — prepared reporting-ready datasets in Snowflake and resolved data-consistency issues feeding Power BI.
- Led a reporting migration for a pharmaceutical client: rebuilt Qlik Cloud reports in Power BI and automated daily distribution to stakeholders via Power Automate.

AMN Healthcare — Medical Interpreter (Part-Time)

Queens, NY

Sep 2025 – Present

- Provide real-time interpretation services in medical and business settings.

AI & DATA ENGINEERING PROJECTS

CMS Medicare Data Pipeline

April 2026 - May 2026

Kafka · Spark · Airflow · AWS S3 · Snowflake

- Built a healthcare data pipeline processing CMS Medicare datasets using **Kafka** for ingestion, **Spark** for distributed transformation, and **Airflow** for orchestration.
 - Designed scalable ingestion and transformation workflows supporting large-volume healthcare datasets and automated scheduling.
 - Implemented data quality validation and dimensional modeling to prepare analytics-ready datasets for downstream reporting.
 - Demonstrated modern data platform architecture combining streaming ingestion, distributed processing, orchestration, and cloud storage.
- [GitHub](#) ↗

Olist Supply Chain & AI Reviews Intelligence

Mar 2026 – Apr 2026

[Python](#) · [Prefect](#) · [Docker](#) · [AWS S3](#) · [Snowflake](#) · [Cortex LLM](#) · [dbt](#) · [Neo4j](#) · [Power BI](#)

- Built an automated ELT pipeline (Python, Prefect, Docker) ingesting **100K+** e-commerce reviews and transactions from AWS S3 into Snowflake at **99.9% reliability**.
 - Engineered multilingual (Portuguese) sentiment analysis in Snowflake using **Cortex LLM** with SQL-based fallback logic for consistent large-scale processing.
 - Applied **Neo4j graph modeling** to surface suspicious review patterns and detect fraud clusters among sellers.
 - Designed a Kimball star schema in dbt with automated data-quality tests; built an executive Power BI dashboard showing the bottom 3% of sellers drove **40%+** of negative sentiment.
 - Linked logistics SLA breaches across 5 regions to a 3.5-point satisfaction drop, enabling targeted interventions.
- [GitHub](#) ↗

US DOT Flights Orchestrated Data Pipeline

Apr 2026 – May 2026

[Apache Airflow](#) · [Docker Compose](#) · [Python](#) · [AWS S3](#) · [Snowflake](#) · [dbt](#)

- Engineered an end-to-end **monthly pipeline** (Dockerized Airflow, Python, AWS S3, Snowflake) ingesting and transforming US DOT (BTS) flight data from raw download to analytics-ready marts.
 - Orchestrated parameterized DAGs — download, GZIP, S3 upload, Snowflake COPY INTO, dbt run/test — with **idempotency checks** at every stage for safe monthly backfills.
 - Established a Snowflake external stage via S3 Storage Integration (IAM trust chain) for serverless loading into raw / staging / core / mart layers.
 - Modeled dbt fact/dim tables (fact_flights, dim_carrier, dim_airport, dim_date with dynamic date spine) using MD5 surrogate keys for join-efficient dimensional modeling.
 - Containerized the full Airflow stack (webserver, scheduler, metadata DB) with Docker Compose for a reproducible, zero-conflict environment.
- [GitHub](#) ↗

NYC Taxi Analytics Engineering Pipeline

Dec 2025 – Mar 2026

[PostgreSQL](#) · [dbt](#) · [Docker](#) · [Power BI](#)

- Built an end-to-end ELT pipeline transforming raw NYC Taxi trip data into analytics-ready reporting tables.
 - Designed layered models across raw / staging / core / mart schemas; implemented data-quality validation for anomalies like unrealistic trip speeds and invalid timestamps.
 - Created daily aggregated marts (trip volume, revenue, duration) and Power BI dashboards for demand, revenue, and airport-traffic analysis.
 - Containerized the dev environment with Docker for a reproducible local warehouse.
- [GitHub](#) ↗

ADDITIONAL EXPERIENCE

Independent Insurance Broker — Client relationships, business development, and regulatory compliance 2016 – 2022

EDUCATION

Stony Brook University — B.S., Electrical Engineering Stony Brook, NY · May 2008

CERTIFICATION

Microsoft Certified: Power BI Data Analyst Associate Expires Jul 2026